

Ex.No: 12	Demonstration of Man-in-the-Middle (MITM) Concept Using Python Socket Programming
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AIM:

ALGORITHM:

CODE:

Code 1 — server.py

```
import socket
HOST = "127.0.0.1"
PORT = 9000
server = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
server.bind((HOST, PORT))
server.listen(1)
print("Test Server started on port 9000")
print("Waiting for MITM relay...")
```

```
conn, address = server.accept()
print("Connection received from:", address)
data = conn.recv(1024)
message = data.decode()
print("Message received at Server:", message)
reply = "Hello Client, message received by Server"
conn.send(reply.encode())
conn.close()
server.close()
```

Code 2 — mitm.py

```
import socket
MITM_HOST = "127.0.0.1"
MITM_PORT = 8000
SERVER_HOST = "127.0.0.1"
SERVER_PORT = 9000
mitm_server = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
mitm_server.bind((MITM_HOST, MITM_PORT))
mitm_server.listen(1)
print("MITM Relay started on port 8000")
print("Waiting for Client...")
client_conn, client_address = mitm_server.accept()
print("\nClient connected:", client_address)
client_data = client_conn.recv(1024)
client_message = client_data.decode()
print("\nMITM observed Client message:")
print(client_message)
server_conn = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
server_conn.connect((SERVER_HOST, SERVER_PORT))
server_conn.send(client_data)
print("\nMessage forwarded to Server")
server_reply = server_conn.recv(1024)
print("\nMITM observed Server reply:")
print(server_reply.decode())
client_conn.send(server_reply)
print("\nReply forwarded to Client")
server_conn.close()
client_conn.close()
mitm_server.close()
print("\nMITM Relay stopped")
```

Code 3 — client-mitm.py

```
import socket
HOST = "127.0.0.1"
PORT = 8000
client = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
client.connect((HOST, PORT))
message = input("Enter message: ")
client.send(message.encode())
```

```
reply = client.recv(1024)
print("Reply from Server:", reply.decode())
client.close()
```

OUTPUT:

```
C:\Users\acer>cd Desktop\MITM_Exercise

C:\Users\acer\Desktop\MITM_Exercise>python server.py
Test Server started on port 9000
Waiting for MITM relay...
Connection received from: ('127.0.0.1', 63713)
Message received at Server: Cryptology Lab
```

```
C:\Users\acer>cd Desktop\MITM_Exercise

C:\Users\acer\Desktop\MITM_Exercise>python mitm.py
MITM Relay started on port 8000
Waiting for Client...

Client connected: ('127.0.0.1', 63711)

MITM observed Client message:
Cryptology Lab

Message forwarded to Server

MITM observed Server reply:
Hello Client, message received by Server

Reply forwarded to Client

MITM Relay stopped

C:\Users\acer\Desktop\MITM_Exercise>
```

```
C:\Users\acer>cd Desktop\MITM_Exercise

C:\Users\acer\Desktop\MITM_Exercise>python client-mitm.py
Enter message: Cryptology Lab
Reply from Server: Hello Client, message received by Server

C:\Users\acer\Desktop\MITM_Exercise>
```

RESULT: